

Gas exchange and Gas Transport & Regulation of respiration

zelalem.banjaw@ju.edu.et

Presentation outline

- ◆ Physical principles of gas exchange and Gas Transport
- ◆ Regulation of respiration

Physical principles of gas exchange

Alveolar Pressure Changes

- ♦ Partial pressure of respiratory gasses: it is a pressure caused by each individual gas

Calculating Partial Pressures

$$P_{\text{gas}} = \text{Fraction of the gas} \times \text{Barometric Pressure}$$

- $PO_2 = 20.93/100 \times 760 \text{ mmHg} = 159 \text{ mmHg}$

- $PCO_2 = 0.03/100 \times 760 \text{ mmHg} = 0.2 \text{ mmHg}$

- $PN_2 = 79.04/100 \times 760 \text{ mmHg} = 601 \text{ mmHg}$

Table: Partial Pressure of Respiratory Gasses (mmHg)

Gas	%(Dry)	Dry Air	Alveolar Air	Venous Blood	Diffusion Gradient
Total	100	760	760	760	
H ₂ O	0.00	0.0	47	47	
O ₂	20.93	159	104	40	64
CO ₂	0.03	0.2	40	47	7
N ₂	79.04	601	573	573	

Partial Pressure Gradients and Gas Solubility

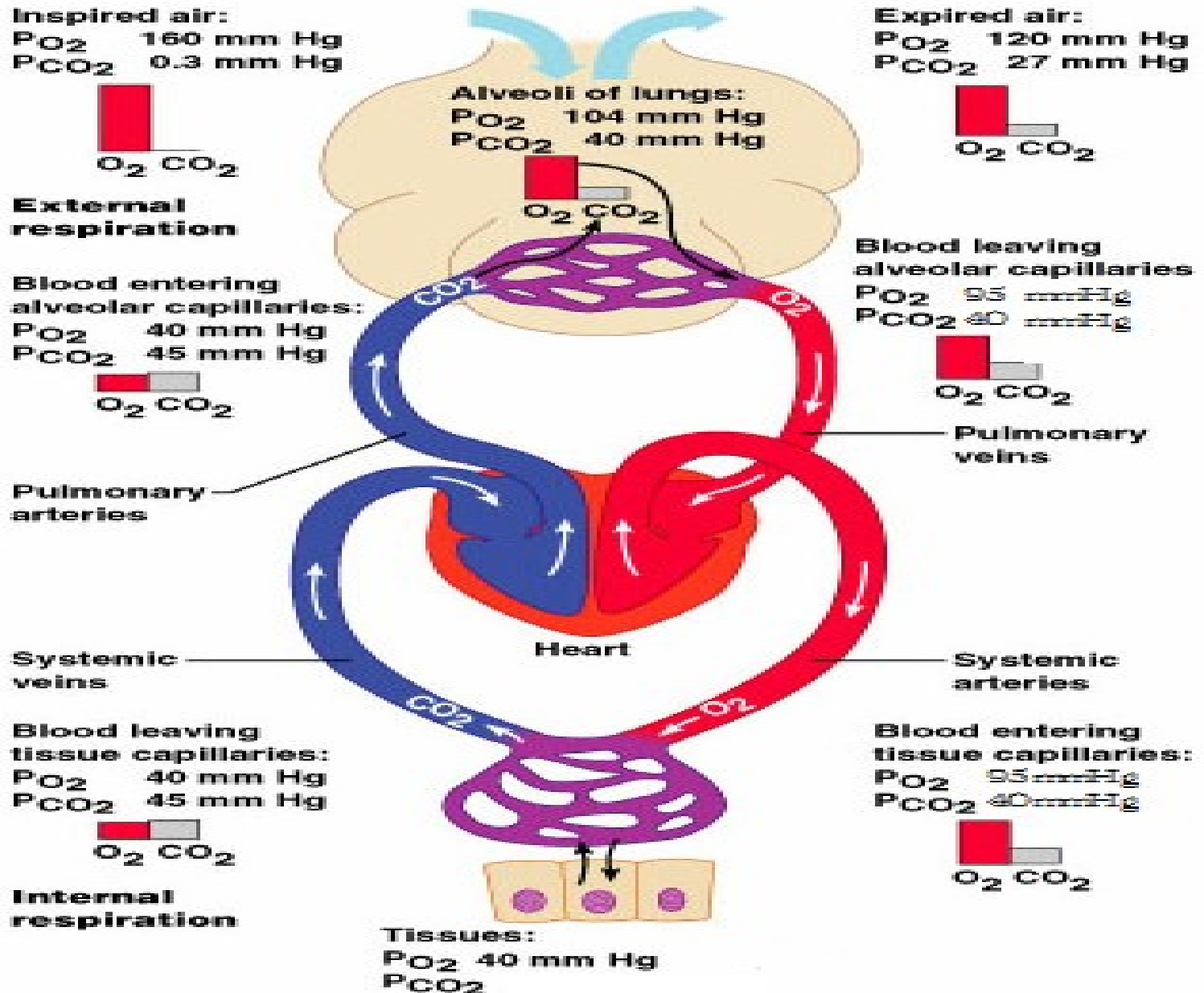
◆ PO_2

- ✓ Capillaries is 40 mmHg
- ✓ Alveoli = 104 mmHg
- ✓ This steep gradient allows oxygen partial pressures to rapidly diffuse from alveoli to capillaries

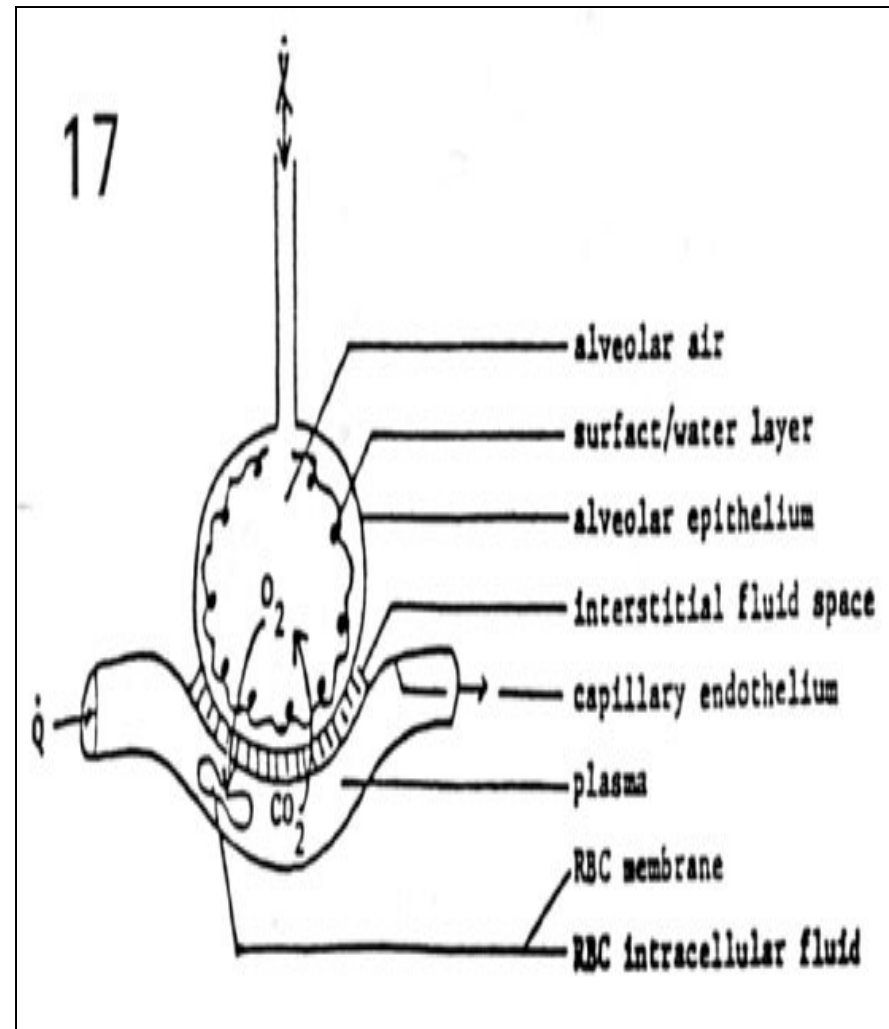
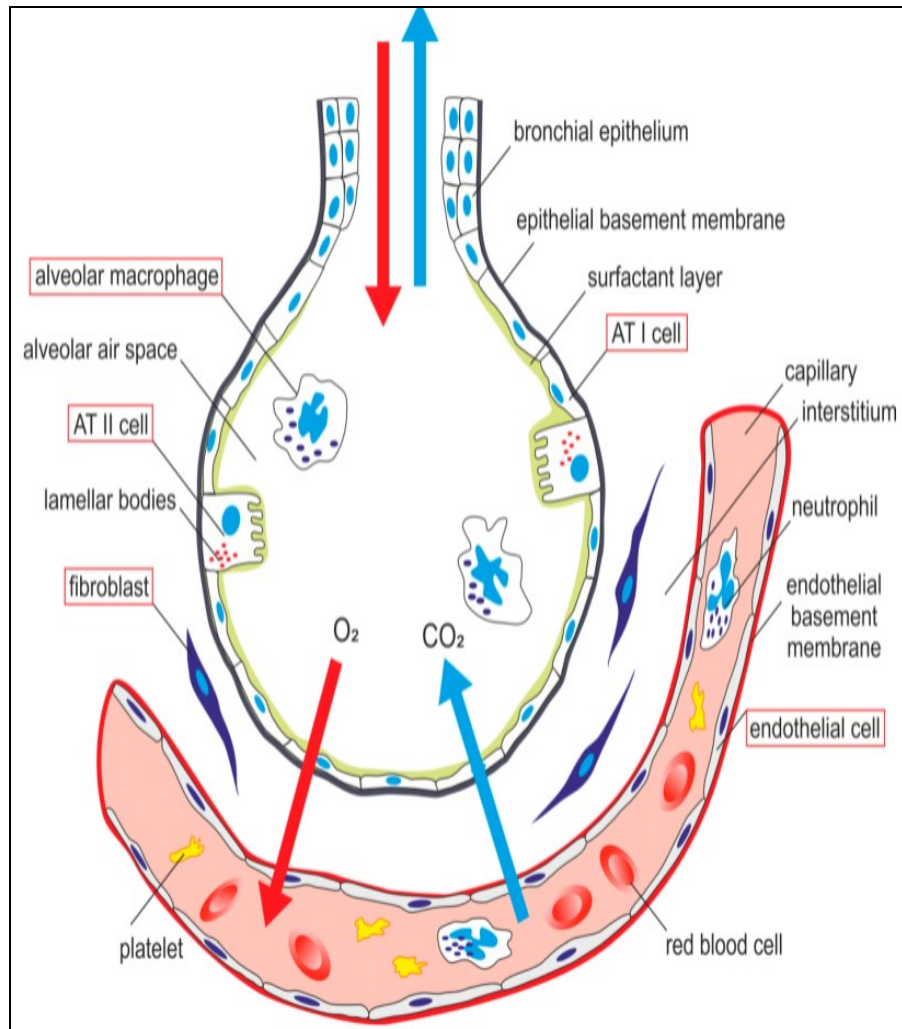
◆ PCO_2

- ✓ Alveolar Capillaries = 46mmHg
- ✓ Alveoli = 40 mmHg
- ✓ has a lower partial pressure gradient
- ✓ It is 20X more soluble in plasma than oxygen as a result it diffuses easily from capillaries to alveoli

Partial Pressure Gradients



Pulmonary Gas Exchange



Factors Affecting Rate of Gas Diffusion

◆ Factors affect rate of gas diffusion through the respiratory membrane

1. Difference in partial pressure on two sides of the respiratory membrane (ΔP)
2. Surface area of the alveolar membrane (A)
3. Solubility of the gas in fluid (S)
4. Membrane Thickness (D)
5. Molecular Weight of the gas (MW)
6. Temperature (T)

$$\text{Rate of diffusion} = \frac{\Delta P \cdot S \cdot A \cdot T}{D \sqrt{MW}}$$

Gas Transport

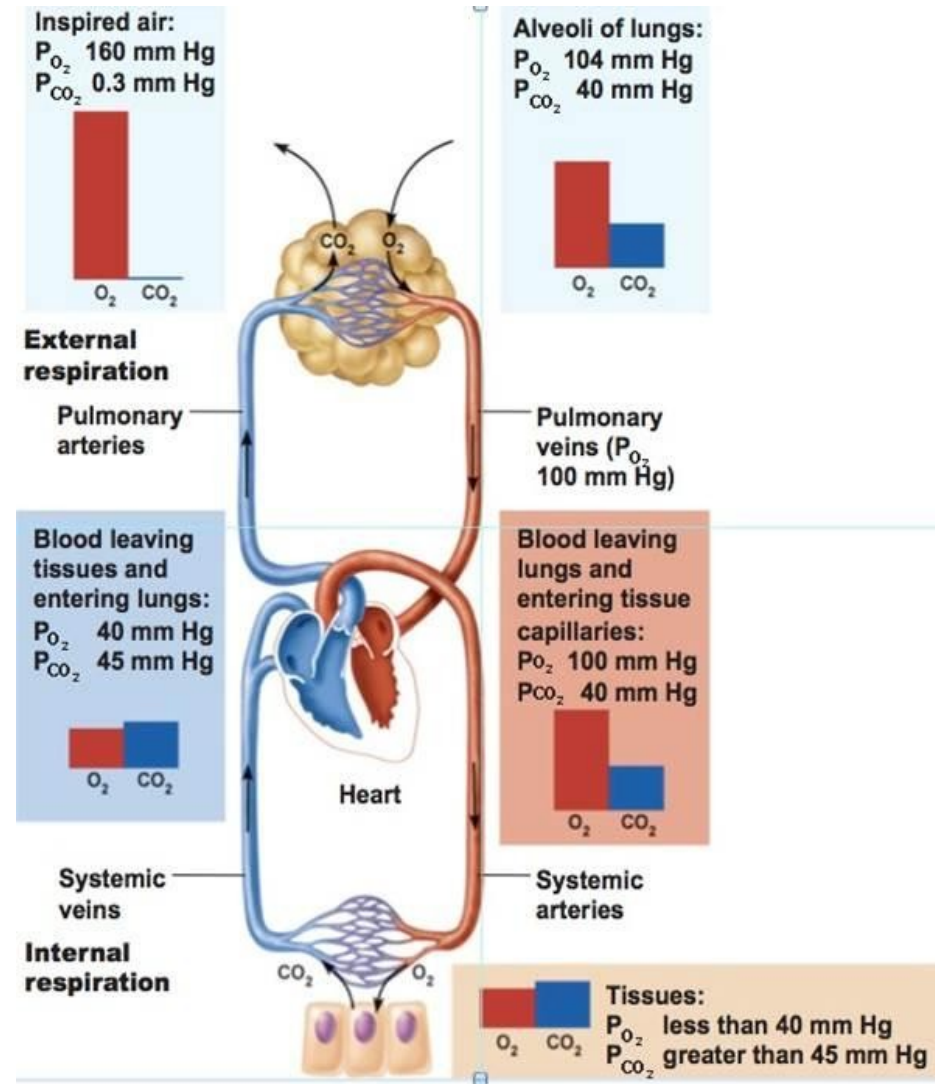
Gas Transport

Transport of Oxygen in Blood

- ✓ In combination with Hb = 97%
- ✓ Dissolved in Plasma = 3%

Transport of CO₂ in Blood

- ✓ In combination with Hb = 23%
- ✓ In form of HCO_3^- = 70%
- ✓ Dissolved in plasma = 7%



Oxygen Transport

Role of Hemoglobin

- ◆ Each Hb molecule binds four oxygen atoms in a rapid and reversible process
- ◆ The hemoglobin-oxygen combination is called **oxy-hemoglobin (HbO₂)**
- ◆ Hemoglobin that has released oxygen is called **reduced hemoglobin (HHb)**

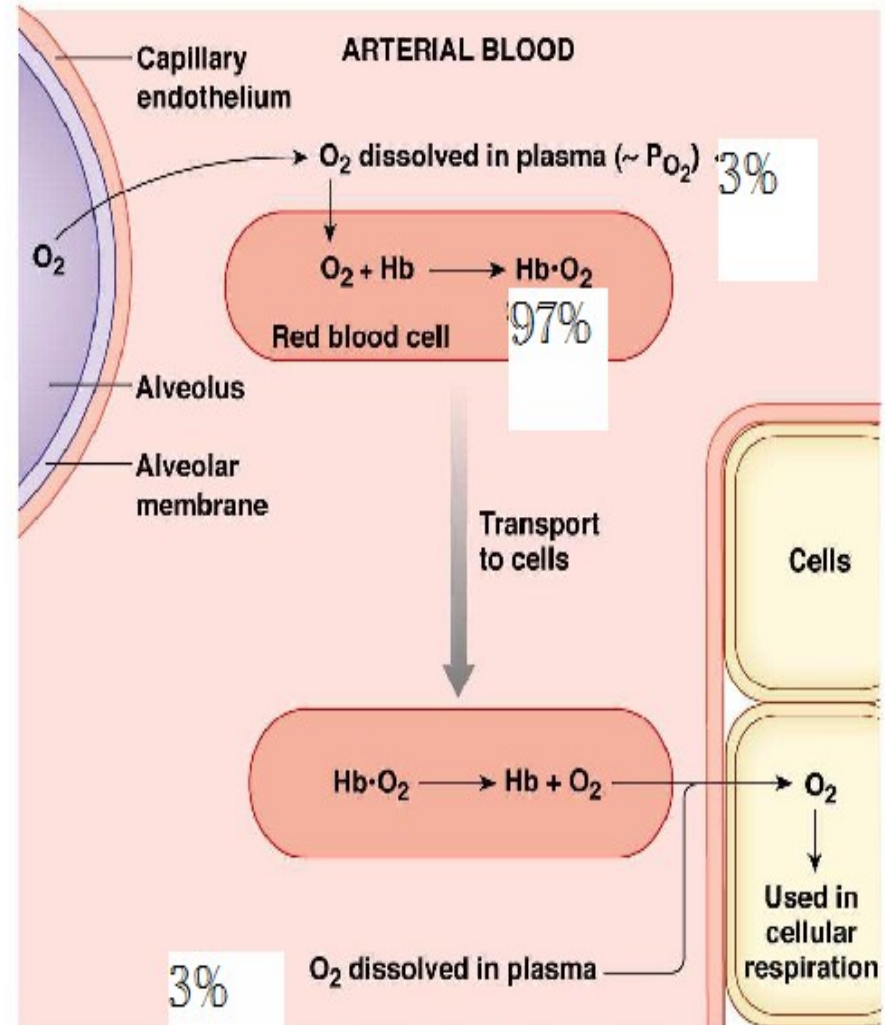
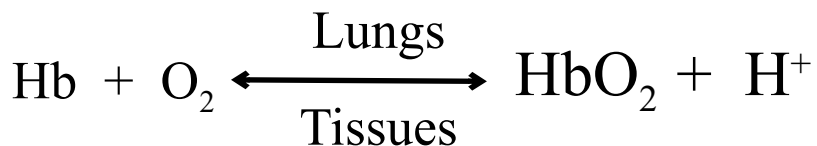


Figure: Transport of O₂ in Blood Mainly in combination with Hb from the Lungs to the Peripheral Tissues down its partial pressure gradient

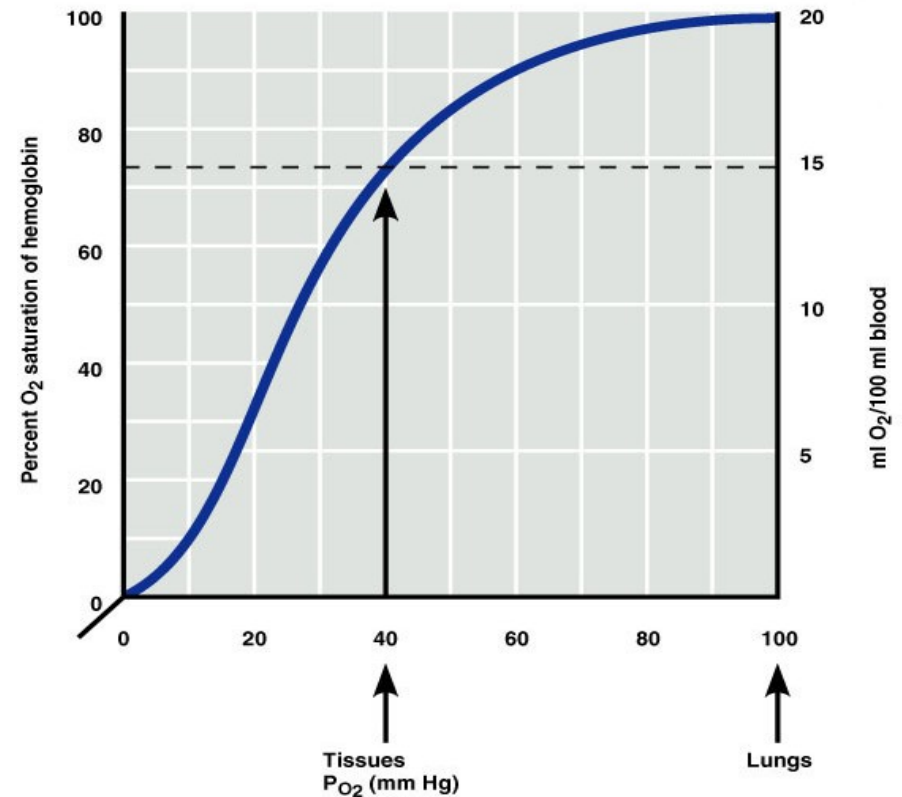
Hemoglobin (Hb)

- ◆ **Saturated hemoglobin:** – when all four hemes of the molecule are bound to oxygen
- ◆ **Partially saturated hemoglobin:** – when one to three hemes are bound to oxygen
- ◆ The rate that hemoglobin binds and releases oxygen is regulated by:
 - ✓ P_{O_2}
 - ✓ Temperature
 - ✓ Blood pH
 - ✓ P_{CO_2} and
 - ✓ The concentration of 2,3 Diphosphoglycerate (DPG) (an organic chemical)

Influence of PO_2 on Hemoglobin Saturation

- ◆ 98% saturated arterial blood contains 20 ml oxygen per 100 ml blood (20 vol. %)
- ◆ As arterial blood flows through capillaries, 5 ml oxygen is released
- ◆ The maximal saturation of Hb in arterial blood (98%) explains why breathing deeply increases has *no effect* on oxygen saturation in hemoglobin

Oxygen-Hemoglobin Saturation Curve



Other Factors Influencing Hemoglobin Saturation

◆ Temperature and allosteric effectors (H^+ , P_{CO_2} , and 2,3-DPG)

✓ All these factors shift O_2 -Hb dissociation curve to the **right down**.

✓ Modify the structure of hemoglobin and alter its affinity for oxygen

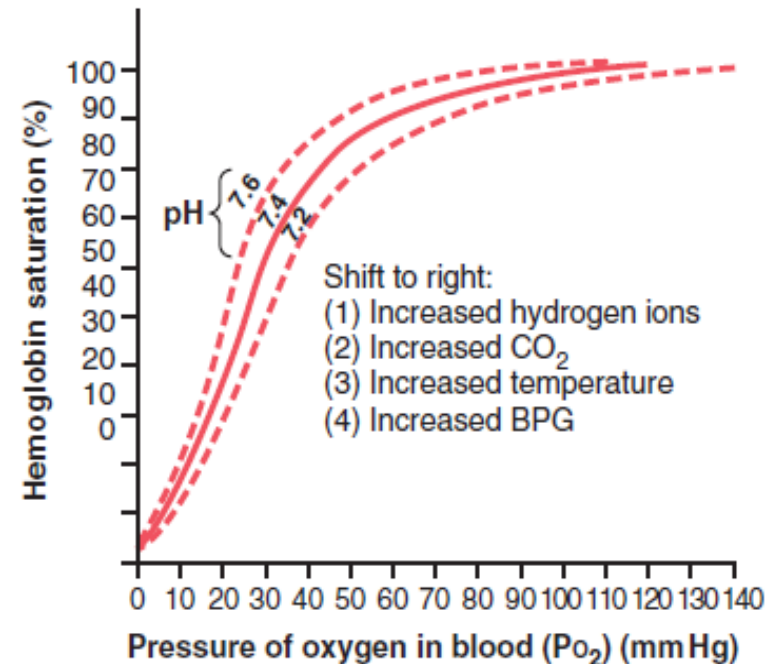
✓ Increases of these factors:

- Decrease hemoglobin's affinity for oxygen (oxygen is less tightly bound to hemoglobin) and

- Enhance oxygen unloading from the blood

◆ Q: What causes left shift?

FACTORS THAT SHIFT THE OXYGEN-HEMOGLOBIN DISSOCIATION CURVE

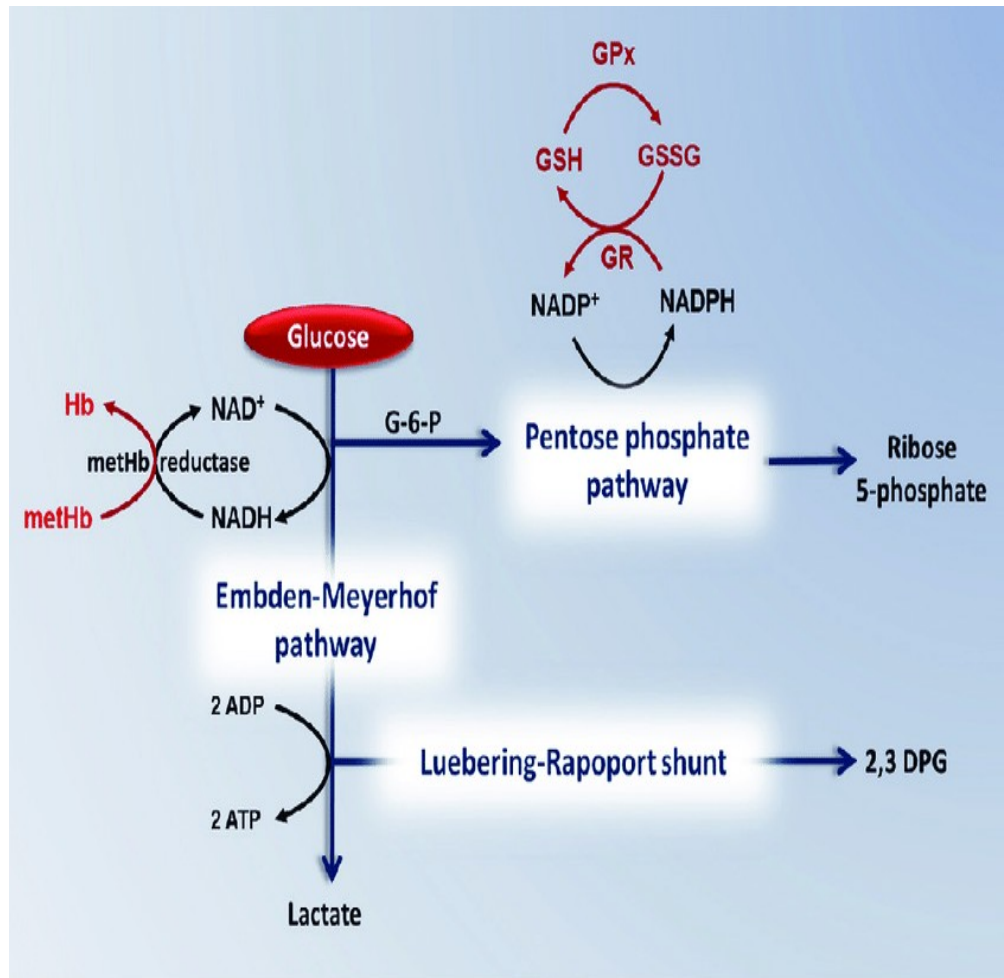


The Bohr Effect:

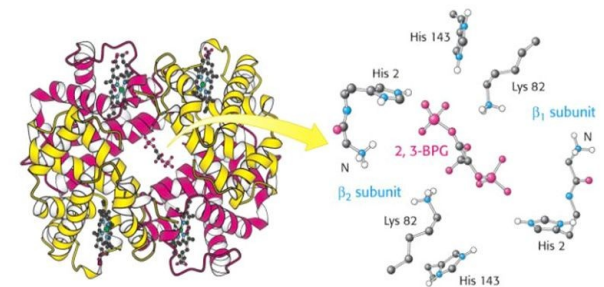
An increase in CO₂ in the blood results in a decrease in hemoglobin association with oxygen. (Oxygen **dissociation** increases).

- 1) As a result of the H⁺ produced by the reaction of CO₂ and water.
- 2) Due to carbaminohemoglobin production.

2,3-biphosphoglycerate



2,3-BPG Binding to Hb



<http://oregonstate.edu/instruction/bb-450/sty/erich10/Slide26.jpg>

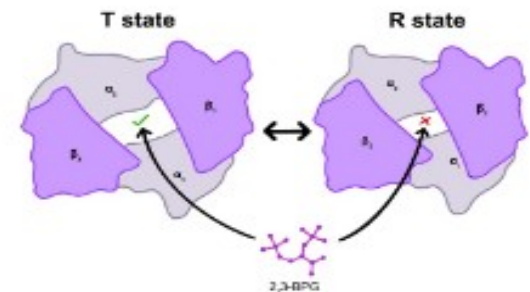


Figure 12. Binding of 2,3-bisphosphoglycerate (BPG) to the T state of Hgb only.

Carbon Dioxide Transport

- ◆ Highest level of Carbon dioxide is formed in the peripheral tissues as the end product of respiration
- ◆ Carbon dioxide is transported in three forms:
 - ✓ bicarbonate -70%
 - ✓ carbaminohemoglobin -23%
 - ✓ Dissolved in plasma-7%

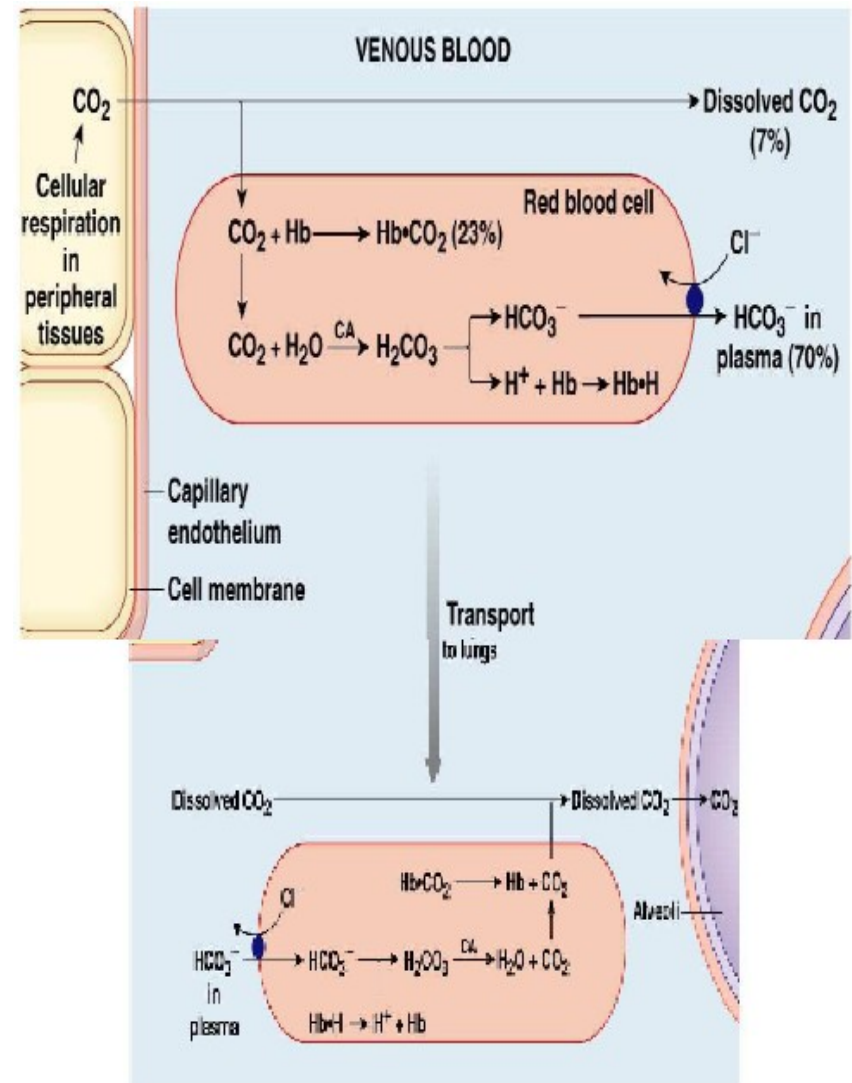


Figure: Carbon Dioxide Transport

Regulation of respiration

Regulation of respiration

- ◆ 2 major respiratory centers in the brain stem

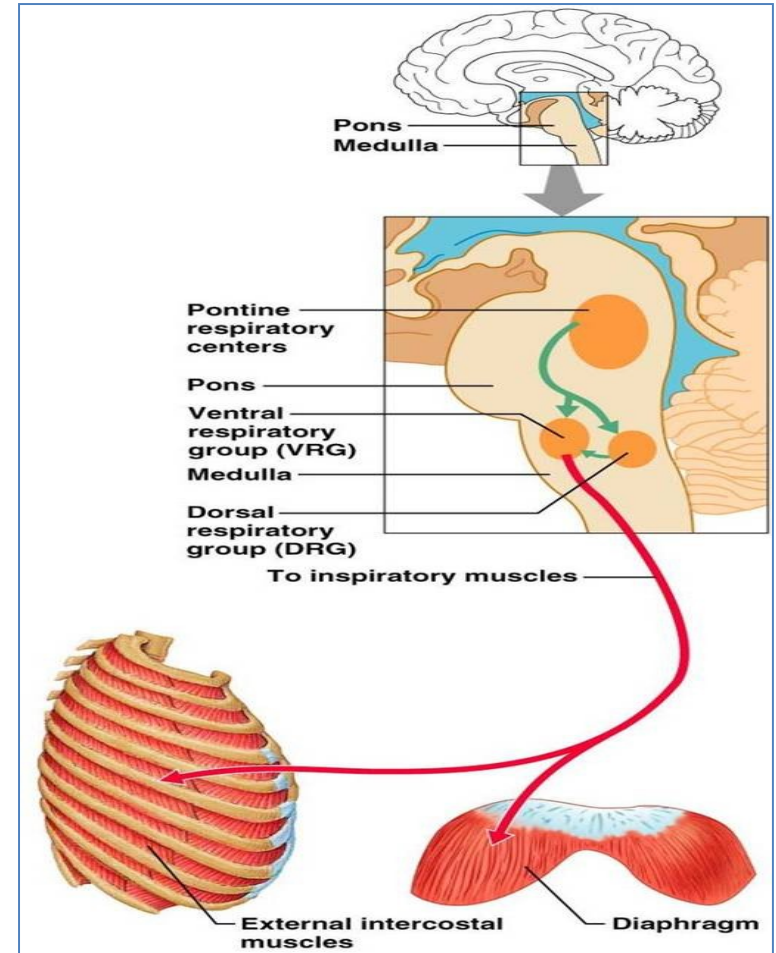
I. Medulla oblongata

a. DRG

b. VRG

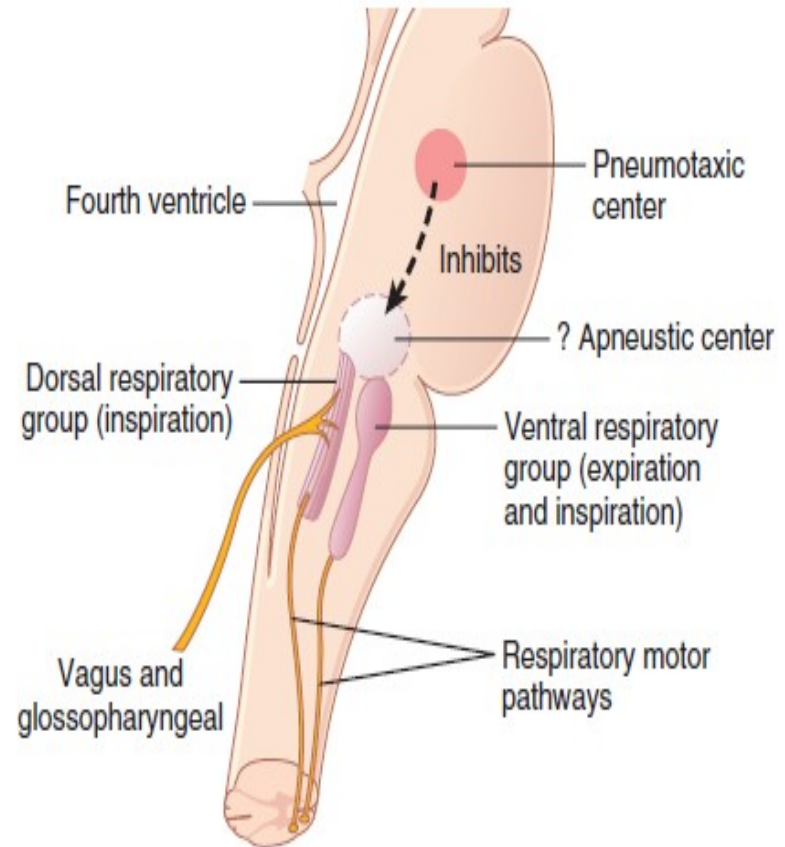
II. Pons

c. Pneumotaxic center



a. Dorsal Respiratory Group of Neurons (DRG)

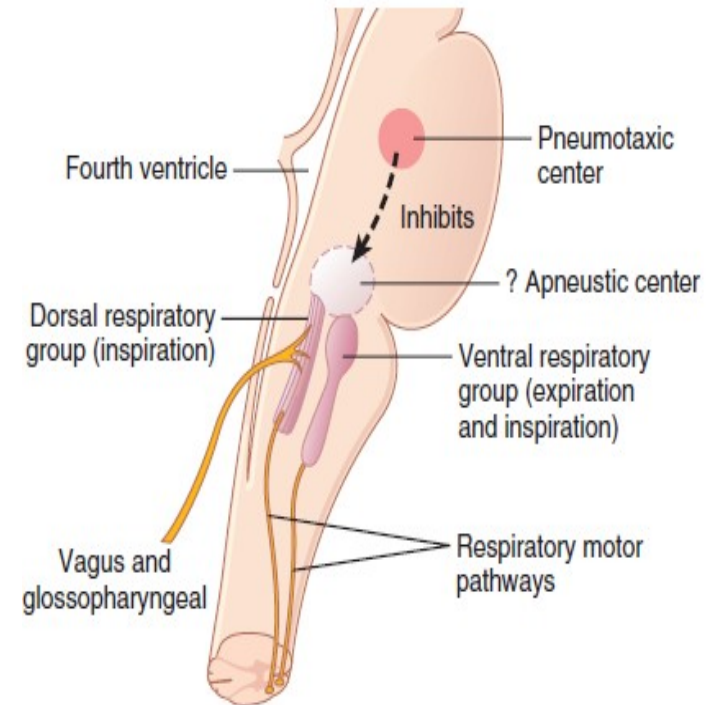
- ◆ Control **Inspiration** and **Respiratory rhythm**
- ◆ Most of its neurons are located within *nucleus of the tractus solitarius* (NTS)
- ◆ Peripheral chemoreceptors, baroreceptors, and several types of receptors in the lungs
→ CN-IX & X → NTS
- ◆ The inspiratory signal is a *ramp signal*.



a. Dorsal Respiratory Group of Neurons (DRG)..

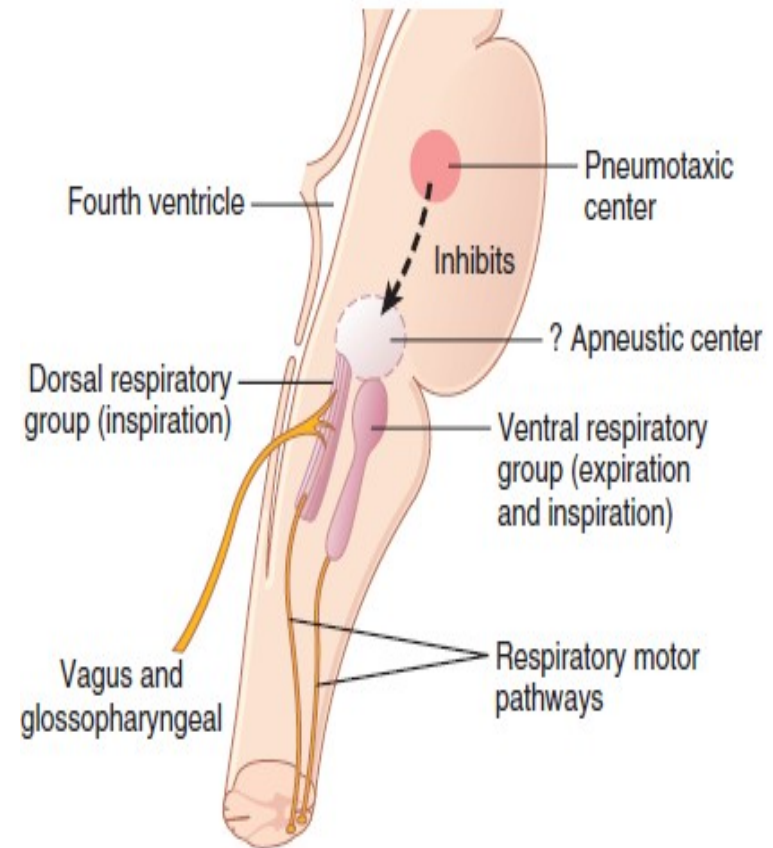
Inspiratory “Ramp” Signal.

- ◆ The nervous signal that is transmitted to the inspiratory muscles, mainly the diaphragm, is **not** an instantaneous burst of action potentials.
- ◆ Instead, it *begins weakly and increases steadily* in a ramp manner for about 2 seconds in normal respiration.
- ◆ It then ceases abruptly for approximately the next 3 seconds, which turns off the excitation of the diaphragm and allows elastic recoil of the lungs and the chest wall to cause expiration.



b. PNEUMOTAXIC CENTER

- ◆ *located dorsally in the nucleus parabrachialis of the upper pons*
- ◆ Control the “switch-off” point of the inspiratory ramp (Limits duration of inspiration)
 - ✓ *strong pneumotaxic signal* → inspiration might last for as little as 0.5 sec → lungs fill only slightly (RR increase)
 - ✓ *weak pneumotaxic signal* → inspiration might continue for 5 sec or more → lungs fill a great excess of air (RR decrease)



c. Ventral Respiratory Group of Neurons

- ◆ Located in each side of the medulla, about 5 millimeters anterior and lateral to the DRG in the *nucleus ambiguus rostrally* and the *nucleus retroambiguus caudally*.
- ◆ Differs from DRG as follows:-
 - i. Remain almost totally *inactive during normal quiet respiration*.
 - ii. Do not participate in the Basic Rhythmical Oscillation
 - iii. Contribute to extra respiratory drive
 - ✓ When the respiratory drive is greater than normal, respiratory signals spill over into VRG from DRG (Heavy exercise)
 - iv. Electrical stimulation of a few of VRG causes inspiration, whereas stimulation of others causes expiration

Hering-Breuer inflation reflex

- ◆ Tidal volume increases to 3X ($>\approx 1.5$ liters per breath) $\rightarrow$ *stretch receptors* in muscular portions of bronchi and bronchioles throughout the lungs $\rightarrow$ *Vagi* $\rightarrow$ *DRG* $\rightarrow$ “switches off” the inspiratory ramp and thus stops further inspiration
- ◆ Protective mechanism for preventing excess lung inflation rather than an important ingredient in normal control of ventilation

CONTROL OF OVERALL RESPIRATORY CENTER ACTIVITY

CHEMICAL CONTROL OF RESPIRATION

- ◆ Control of ventilation by PO_2 , PCO_2 and H^+ is achieved via chemo-receptors

I. Central chemo-receptors

- ✓ located on the *ventral surface of the medulla*

- ✓ PCO_2 and H^+

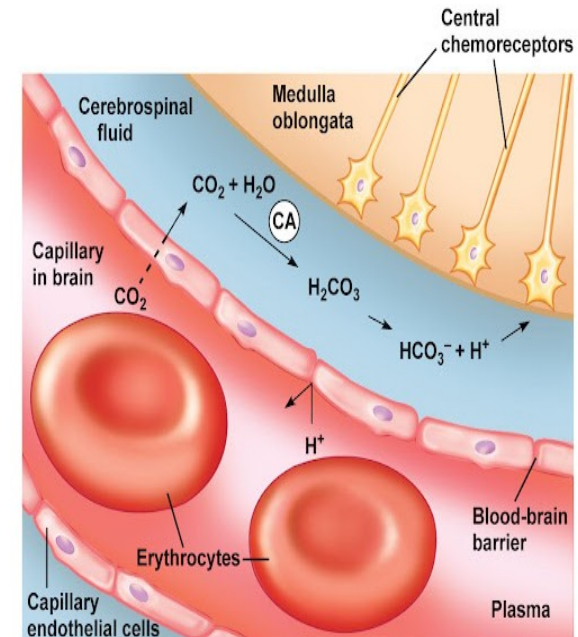
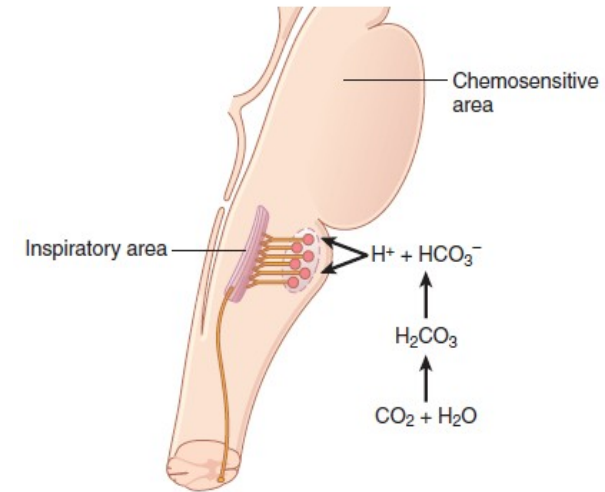
II. Peripheral chemo-receptors

- ✓ located in the *carotid bodies* and *aortic bodies*

- ✓ PO_2

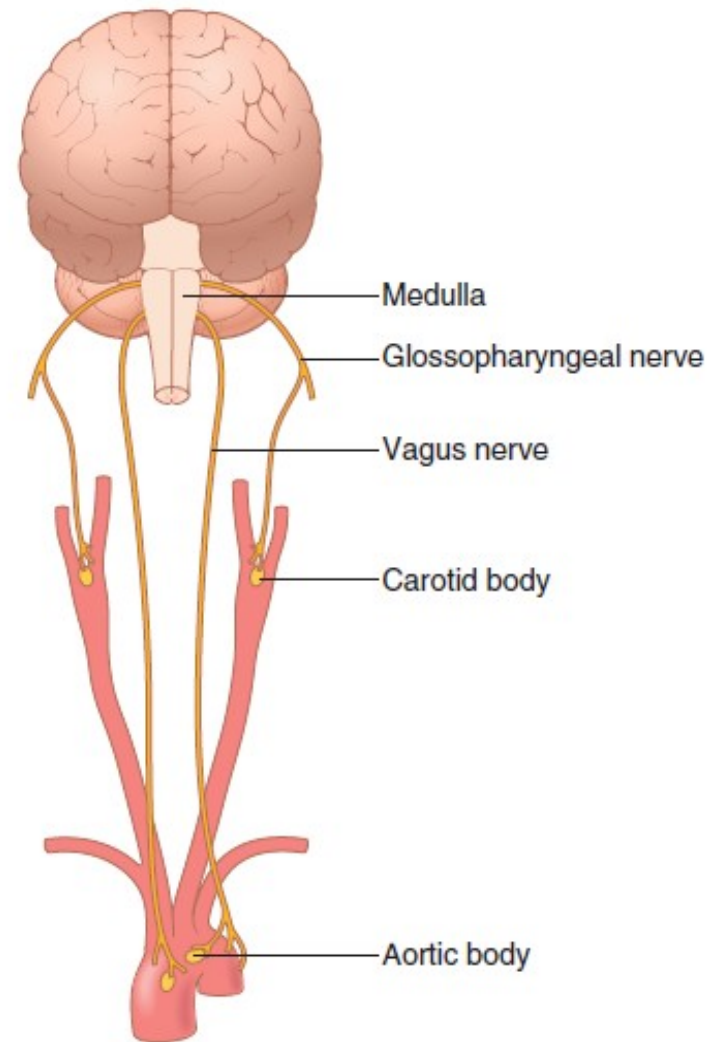
i. Central chemo-receptors system

- ◆ **Most important** regulator of ventilation
- ◆ Central chemo-receptors are found in the **medulla**.
- ◆ CO_2 has *little direct effect* in stimulating the neurons in **chemo-sensitive area**, it does have a *potent indirect effect*
- ◆ Respond to changes in **CO_2 -induced** production of H^+ in CSF
- ◆ **Excitation of the Chemosensitive Neurons by Hydrogen Ions Is Likely the Primary Stimulus**
- ◆ *Blood-brain barrier allows the diffusion of CO_2 but is impermeable to H^+ (i.e. why CO_2 have a more potent effect in stimulating the chemosensitive neurons)*



ii. Peripheral chemoreceptor system

- ◆ Important for detecting changes in O_2 in the blood
- ◆ Respond to a lesser extent to changes in CO_2 and H^+ concentrations.
- ◆ Most of the chemo-receptors are in the *carotid bodies-glomus cells sense low O_2* (Afferent: glossopharyngeal nerves).
- ◆ Few are also in the *aortic bodies* (Afferent: Vagi)
- ◆ Peripheral O_2 chemoreceptors do not contribute in regulating normal ventilation unless arterial PO_2 falls below 60 mm Hg



Phenomenon of “Acclimatization”

- ◆ Slow Mountain climbers (over a period of days) breathe much more deeply and therefore can withstand lower atmospheric O₂ concentrations than fast

climbers (period of hours). **Why ?**

- ◆ Acclimatization to hypoxia includes
 - ✓ Increase in pulmonary ventilation
 - ✓ Increase in RBC
 - ✓ Increase in diffusion capacity of the lungs,
 - ✓ Increase in vascularity of the tissues
 - ✓ Increase in the ability of the cells to use available oxygen.

Q: Which of the following sets of changes best describes a Himalayan native living in the Himalayas, compared to a sea-level native living at sea level?

	Hematocrit	Arterial PO ₂	Arterial O ₂ content
A)	Decreased	Decreased	Decreased
B)	Decreased	Decreased	No difference
C)	Decreased	Increased	Decreased
D)	Decreased	Increased	No difference
E)	Increased	Decreased	Decreased
F)	Increased	Increased	Decreased
G)	Increased	Increased	No difference
H)	Increased	Decreased	No difference

Thank you